



INSURPRIME: CAN YOU GUESS THE INSURANCE PREMIUM?

by **Crédit Agricole Assurances**

March 1, 2026

Nelson Tejodonkeng & Kenza Chaabouni



CONTENTS

1	Introduction	3
2	Preliminaries	3
2.1	General Setup	3
2.2	Target Definition	3
2.3	Metric	4
3	Exploratory Data Analysis & Feature Engineering	4
3.1	Dataset Overview	4
3.2	Exploratory Data Analysis & Data Cleaning	4
3.3	Feature Engineering	5
4	Methodology & Modeling Strategy	5
4.1	Baseline Generalized Linear Model	6
4.1.1	Model Diagnostics	6
4.1.2	Synthesis of Model Performance	8
4.2	Histogram-based Gradient Boosting Regression	8
4.2.1	Model Diagnostics	9
4.2.2	Tuned Model	10
4.2.3	Modified Model	11
4.2.4	Synthesis of Model Performance	11
4.3	CatBoost	11
4.3.1	Model Diagnostics	12
4.3.2	Tuned CatBoost Model	14
4.3.3	Analysis of Tuning Stability	14
4.3.4	Synthesis of Model Performance	14
5	Model Selection and Evaluation	15
6	Conclusion	15

1

INTRODUCTION

Accurately pricing insurance contracts is a central problem in actuarial science, as premiums must balance risk adequacy, competitiveness, and solvency constraints. In non-life insurance, and particularly in agricultural insurance, losses are often rare, highly skewed, and heavy-tailed, making prediction challenging due to class imbalance. Fire risk, which represents a substantial share of claims in multi-risk agricultural contracts, is especially challenging due to its low frequency and potentially catastrophic severity. Therefore, improving the modeling of such risks is crucial for insurers' financial stability while offering fair premiums[3].

In this report, we present our solution to the challenge [1] offered by Crédit Agricole Assurances, aiming to predict the pure fire premium for agricultural insurance contracts. The premium is decomposed into claim frequency, average claim severity, and exposure. The classical approach to insurance pricing relies on Generalized Linear Models (GLM), which remain a benchmark. Frequency is commonly modeled using Poisson or Negative Binomial distributions, while severity is modeled using Gamma or Tweedie distributions. The frequency–severity decomposition has been shown to be effective and remains a cornerstone of actuarial modeling. More recently, tree-based models have demonstrated superior predictive performance on complex, high-dimensional insurance data[4]. Gradient boosting methods[2], including histogram-based gradient boosting and CatBoost, have been shown to outperform GLMs in terms of predictive accuracy. In particular, CatBoost has proven effective for insurance data due to its native handling of categorical variables and robustness to target leakage.

In this report, we offer three modeling approaches to this challenge. Mainly, a baseline GLM model, a Histogram-based Gradient Boosting Regressor, and a CatBoost regressor. The report proceeds as follows: Section 2 presents the preliminaries; Section 3 describes the data and the relevant feature engineering; Section 4 details the used models and their results; Section 5 presents the evaluation and the model selection and Section 6 concludes.

2

PRELIMINARIES

2.1 GENERAL SETUP

We are given a training dataset¹ containing agricultural insurance contracts with information concerning the buildings, surface, capital data ect. Our task is to predict the pure fire premium for each contract, which depends on the frequency and severity of claims. The challenge requires building a model that can accurately estimate premiums for unseen policies while handling rare, heavy-tailed fire losses.

2.2 TARGET DEFINITION

In this challenge our target is the pure fire premium (charge), defined as:

$$CHARGE_i = \widehat{FREQ}_i * \widehat{CM}_i * Year$$

where $\widehat{FREQ}_i$ is the predicted claim frequency, $\widehat{CM}_i$ the predicted average claim cost (severity), and $Year$ the exposure in years for contract i .

¹The dataset and further information on the challenge can be found here

2.3 METRIC

Model performance is assessed using the Root Mean Square Error (RMSE) on the charge:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (CHARGE_i - \widehat{CHARGE}_i)^2}$$

where $CHARGE_i$ denotes the observed charge and $\widehat{CHARGE}_i$ the predicted charge.

3

EXPLORATORY DATA ANALYSIS & FEATURE ENGINEERING

3.1 DATASET OVERVIEW

The training dataset has 383610 rows and 374 features, taking 1.1+ GB memory usage. We have 280 categorical features and 94 numerical ones. The columns include the insured activity, the guarantee subscription indicators, number of buildings, employees, and claims declared at subscription, surface data, capital data, and prevention data. Our target variable is the pure fire premium (CHARGE) obtained by multiplying the severity (CM), the frequency (FREQ), and the Year (ANNEE-ASSURANCE).

3.2 EXPLORATORY DATA ANALYSIS & DATA CLEANING

The dataset shows extreme zero inflation, with at least 99% of FREQ, CM, and CHARGE being zero, and a few very large claims dominating the variance, indicating rare events. This is a highly skewed loss distribution.

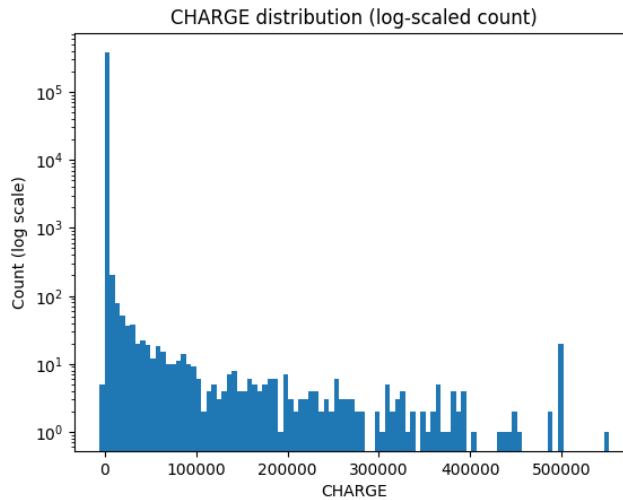


Figure 1: Charge distribution with log-scale count

The CHARGE distribution confirms the zero-inflation, with the vast majority of policies having no fire claims, and a long right tail with a small number of very large losses. The heavy-tailed structure highlights why modeling CHARGE directly is difficult and hence a two-part frequency-severity approach is necessary.

In addition, the dataset is highly sparse in a structured way, we observe a lot of missing data, with 183 columns having more than 50% missing data. This is probably the case because these columns describe a specific type of building or optional clauses which are not applicable for the whole dataset. The missing values are likely useful and informative in this case and should therefore not be treated as noise. Hence, we keep the missing data as it is informative, but drop the columns with more than 95% missing values. Moreover, several numeric variables, such as surface, capital, and risk, contain sentinel values (e.g., 1000, -1, -3) representing “not applicable” or errors, which we replaced with NaN.

We observe that many numerical variables are actually categorical classes. Particularly, variables with two unique counts likely encode binary categorical classes (0/1) and not continuous ones. Thus, these binary indicators are converted to categorical, and object columns are converted to categories for memory efficiency. In addition, some numerical variables have a low cardinality. For example, the column Risk takes a small number of discrete integer levels, rather than a smooth continuous range, indicating they are risk classes or rating bands, not numeric expectations. Therefore, their numeric spacing has no true quantitative meaning, so they are treated as categorical variables. Thus, low-cardinality numeric variables and binary indicators are converted to categorical, and object columns are converted to categories for memory efficiency.

Columns with a single unique value are removed, and highly multicollinear columns, including certain buildings and equipments variables, are dropped to reduce redundancy.

Finally, negative target values in CM and CHARGE are clipped to zero, and the year was used as an offset in the frequency model, while the separate modeling of frequency and severity accounts for the rare-event and heavy-tail structure of the claims.

3.3 FEATURE ENGINEERING

Based on our EDA observations, the feature engineering focuses on summarizing and aggregating sparse and coded numeric variables, capturing farm complexity, and incorporating environmental and risk information.

We created aggregate features for surface variables (total, max, and count of non-zero surfaces) and building variables (total buildings, average surface per building). We then made a simple complexity proxy index reflecting farm complexity based on the sum of buildings and surface.

Similarly, capital values were aggregated and log-transformed to summarize insured capital. In addition, prevention measures were counted and a flag for no prevention was added.

Weather and environmental risks were encoded as derived indices, and rare categories were grouped to improve stability. Moreover, a new-business flag was also derived from the seniority column and risk-related columns were summarized as alert counts. Finally, simple interactions were introduced to capture nonlinear relationships relevant for claim frequency and severity.

We note that in this challenge, data cleaning was crucial for modeling, as the dataset contained many sentinel values, structured missing data, and numeric columns that actually encoded categorical information. Properly handling these issues, by replacing sentinels with NaN, dropping ultra-sparse or constant columns, converting coded numerics and binaries to categorical, and clipping negative targets, was essential to ensure that models could learn meaningful patterns without being misled by noise or artificial values.

4

METHODOLOGY & MODELING STRATEGY

To solve this problem, we built three models: a Generalized Linear Model (GLM), a Histogram-based Gradient Boosting Regressor (HGBR), and a CatBoost model. We incorporated the year (ANNEE-ASSURANCE) as

an exposure offset in the frequency models, and we evaluated all predictions using RMSE to measure accuracy. This approach allows us to compare GLM and tree-based methods on the rare-event, heavy-tailed insurance data.

4.1 BASELINE GENERALIZED LINEAR MODEL

We implement the benchmark **Generalized Linear Model** (GLM) described in the challenge [1]. This benchmark models insurance charge through a two-part structure: a Poisson GLM with a log link for claim frequency, and a Tweedie GLM with a log link for average claim cost, whose product (multiplied by exposure) yields the predicted charge.

The frequency component uses a Poisson distribution with log link, while the severity component uses a Tweedie distribution with power parameter $p = 1.5$, log link, and mild $L2$ regularization ($\alpha = 0.1$).

A key advantage of this GLM benchmark is its computational efficiency: our implementation trains in approximately 1.6 seconds on the full dataset, substantially faster than tree-based approaches.

We obtain a validation RMSE of 7043.09, and a relatively good test RMSE of 5603.60². This makes GLM a solid baseline despite its simplicity.

4.1.1 • MODEL DIAGNOSTICS

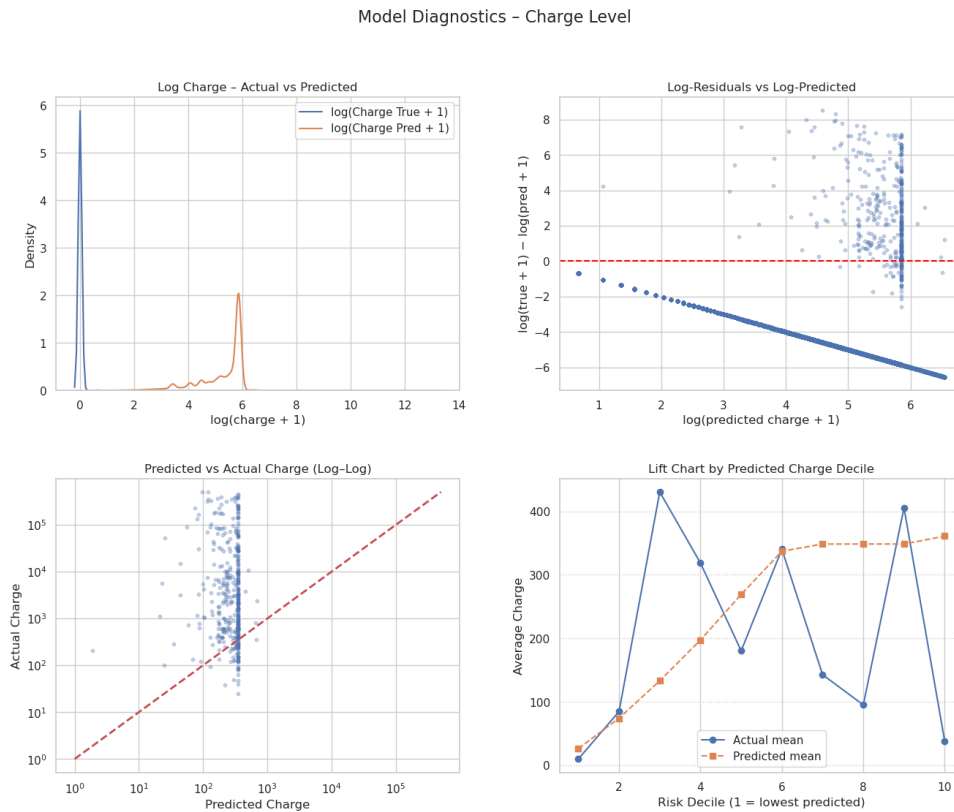


Figure 2: Model diagnostics at the Charge level using the GLM model

²We note that the validation RMSE are quite high across all models, while the test RMSE are much better. The high validation RMSE probably arises from random variation from a few large claims falling into the validation splits, which makes the metric very unstable in rare-event, heavy-tailed data. The test RMSE across models beats the benchmark, meaning the models actually generalize well and are performing strongly.

We analyse the baseline model performance, based on the validation set predictions, at the charge level by plotting the following figures:

- **Log Charge – Actual vs Predicted:** The predicted log-charge distribution is far higher and narrower than the true distribution, showing that the model systematically overestimates charge and fails to represent real variability.
- **Log-Residuals vs Log-Predicted:** Residuals show a strong downward trend, showing that the model overpredicts (increasingly) charge as predicted values rise.
- **Predicted vs Actual Charge (Log-Log):** Predicted charges cluster tightly around a narrow band while actual charges span several orders of magnitude. The model underfits the true charge distribution.
- **Lift Chart by Predicted Charge Decile:** The predicted charges increase with deciles while actual charges fluctuate wildly. The model captures ranking weakly.

Model Diagnostics – Frequency & Severity

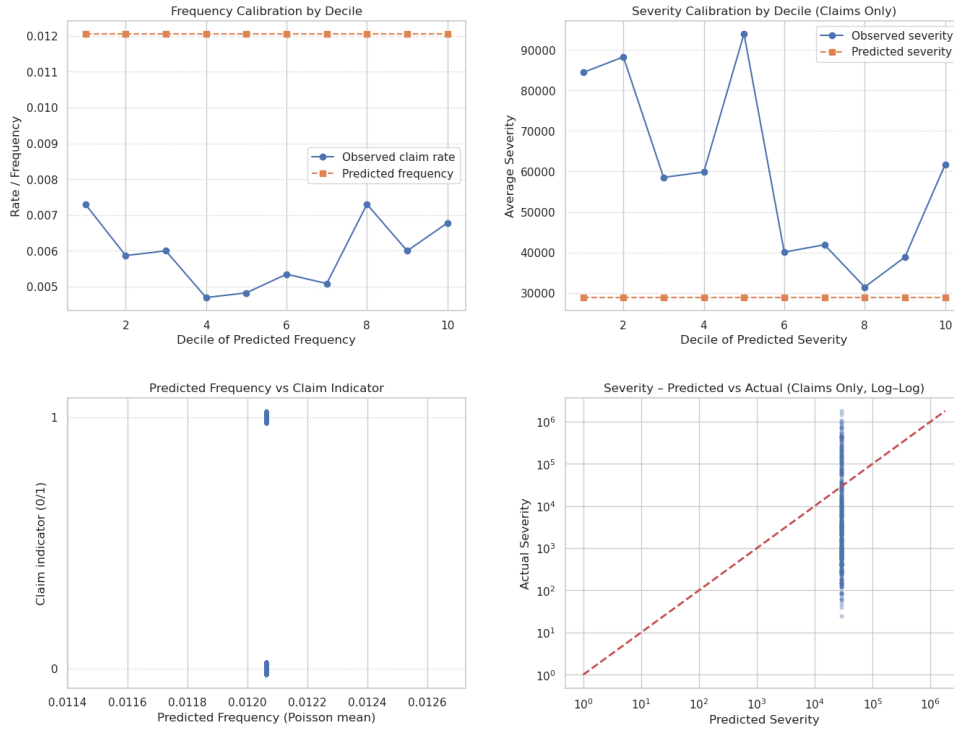


Figure 3: Model diagnostics at the Frequency and Severity level using the GLM model

We analyse the baseline model performance, based on the validation set predictions, at the frequency and severity level by plotting the following figures:

- **Frequency Calibration by Decile:** The predicted frequencies are nearly constant across all deciles, and so the Poisson GLM is not capturing meaningful variation in claim likelihood.
- **Severity Calibration by Decile:** The severity model underfits severities and shows almost no differentiation across deciles.
- **Predicted Frequency vs Claim Indicator:** Predicted frequencies are in a single narrow band. The model outputs nearly identical values regardless of claim outcome.

- **Severity – Predicted vs Actual (Claims Only, Log–Log):** Predicted severities form a vertical line far below the identity line, showing that the model predicts a constant severity and fails to capture the heavy-tailed actual distribution.

While the GLM model achieves a relatively good test RMSE (5603.60), diagnostic plots, on the validation outcomes, reveal that it underfits the data and produces nearly constant predictions.

Furthermore, we attempted to tune the GLM benchmark by varying the hyperparameters of both components, such as the Poisson regularization strength for frequency and the Tweedie power and regularization for severity. With all combinations, the validation RMSE remained strictly constant at 7043.09, indicating that none of the parameter changes altered the model’s predictions. The model collapses to a near-constant solution, so changing hyperparameters does not change predictions at all.

4.1.2 • SYNTHESIS OF MODEL PERFORMANCE

The GLM provides a fast benchmark but systematically underfits the data. Diagnostics show that both the Poisson frequency and Tweedie severity components collapse toward nearly constant predictions, failing to reflect the true variability of rare and heavy-tailed fire claims. While RMSE appears competitive, this is largely due to the dominance of zero charges: the model offers limited risk differentiation, weak calibration across deciles, and consistently biased residual patterns. Overall, the GLM highlights the limitations of linear models for high-dimensional, zero-inflated agricultural insurance data.

4.2 HISTOGRAM-BASED GRADIENT BOOSTING REGRESSION

Histogram-based Gradient Boosting Regression (HGBR) is a tree-based ensemble method. First, the continuous features are binned into discrete histograms, which makes split-finding much faster and more memory-efficient. This is quite crucial given the size of our dataset after cleaning and feature engineering, (383610, 380), with a memory usage of 1.1+ GB. Trees are then added sequentially, each one fitting the residuals of the previous model to minimize a chosen loss. Importantly, HGBR natively handles missing values by treating NaNs as a separate category during node splitting. This property is particularly valuable in our challenge, where missing values are both frequent and informative; consequently, imputing them would likely degrade model performance.

Our pricing model follows a two-part (frequency–severity) structure implemented with HGBR.

- **Frequency model:** Claim counts are modeled using Poisson loss and a log link, which works well for non-negative count data. The year is used as exposure and is incorporated as a sample weight, which ensures that the claim rates are learned per year. Regularization is enforced via limited tree complexity and L2 penalties.
- **Severity model:** Conditional on having at least one claim, average claim cost is modeled using Gamma loss, which is a right fit for right-skewed distributed monetary amounts. Training is restricted to observations with positive charges, predictions are constrained to be positive, and extreme values are clipped to reduce sensitivity to outliers.

Models are trained with a low learning rate and many boosting iterations, and final predictions are stabilized using bagging over multiple random seeds to reduce variance. The final predicted premium is computed as a function of the predicted frequency, predicted severity and exposure. We obtain a validation RMSE of 7041.14, and a competitive test RMSE of 5598.75, which is the best performance among the models we evaluated.

4.2.1 • MODEL DIAGNOSTICS

Model Diagnostics – Charge Level

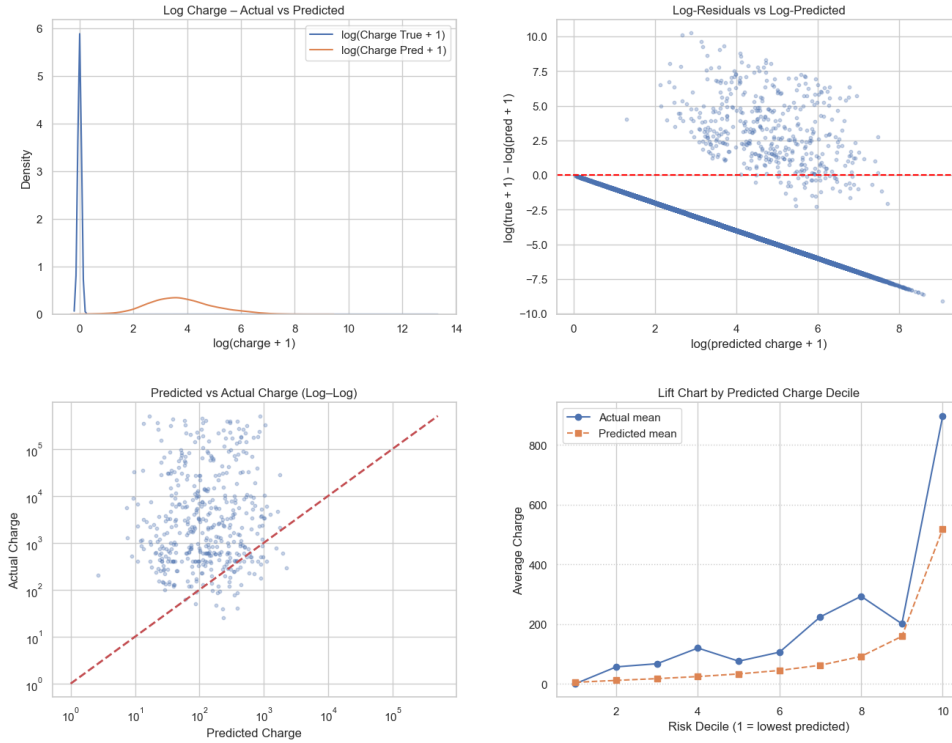


Figure 4: Model diagnostics at the Charge level using the HGBR model

We analyse our model’s performance, based on the validation set predictions, at the charge level by plotting the following figures:

- **Log Charge – Actual vs Predicted:** Actual log-charges are extremely concentrated near zero (as seen in EDA) because most policies have no claims. However, predicted log-charges show a wider and smoother distribution. This suggests the model spreads risk more evenly than the true data.
- **Log-Residuals vs Log-Predicted:** Residuals become increasingly negative as predicted charge increases so the model under-predicts high charges. The horizontal line at 0 shows unbiased predictions would scatter symmetrically around it but here they do not. Therefore, we observe that the model is not well calibrated at the high end.
- **Predicted vs Actual Charge (Log-Log):** Most large actual claims lie above the diagonal so the model systematically underestimates big losses. Predictions cluster in mid-range while actual charges span several orders of magnitude. Heavy-tail behavior is not fully captured by the current model.
- **Lift Chart by Predicted Charge Decile:** Rising actual mean across deciles shows the model can differentiate low- vs high-risk policies. But predicted means grow much more slowly, so the model under-ranks the highest-risk group. Decile 10 gap confirms underestimation of the very top risks.

Model Diagnostics – Frequency & Severity

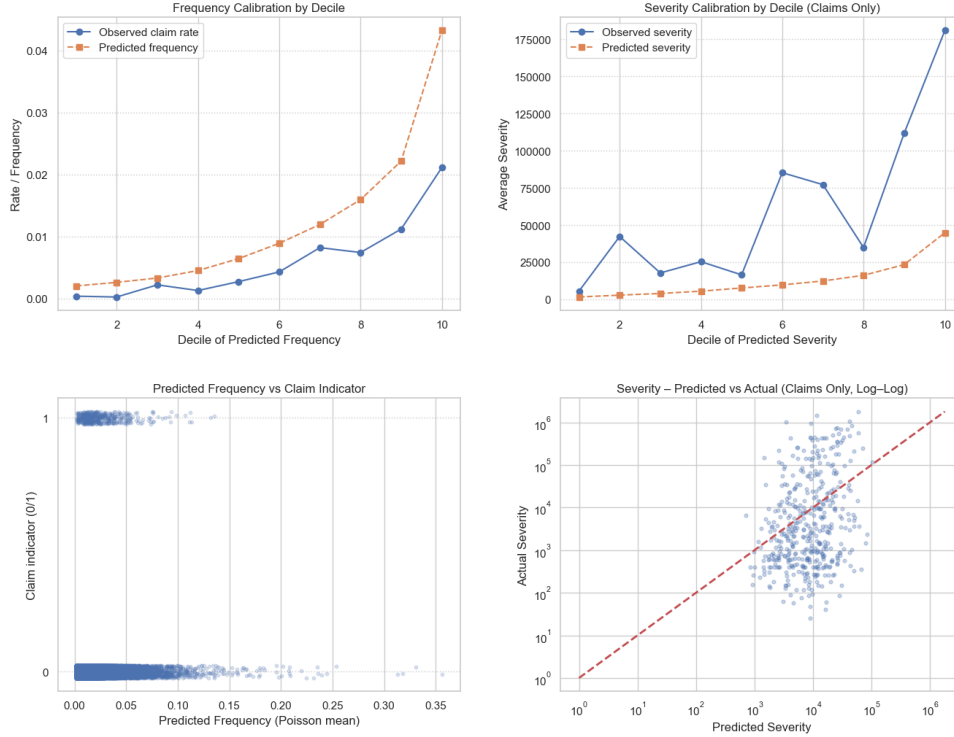


Figure 5: Model diagnostics at the Frequency and Severity level using the HGBR model

We analyse our model’s performance, based on the validation set predictions, at the frequency and severity level by plotting the following figures:

- **Frequency Calibration by Decile:** Predicted frequency increases smoothly across deciles, showing good ranking ability. However, predicted rates are consistently higher than observed claim rates. This indicates systematic overestimation of frequency, especially in the top deciles.
- **Severity Calibration by Decile:** Observed severity is very noisy and jumps sharply in the top decile, reflecting extreme heavy-tailed claims. Predicted severity grows smoothly but stays far below actual values, especially in high-risk groups. The model is underestimating large losses and lacks tail calibration.
- **Predicted Frequency vs Claim Indicator:** Most predicted frequencies are near zero, consistent with the extremely rare-event nature of fire claims. Claim cases (1’s) vertically align at low predicted frequencies so the model fails to separate claim vs non-claim. The frequency model barely distinguishes risky farms from normal ones.
- **Severity – Predicted vs Actual (Claims Only, Log–Log):** Points mostly sit above the diagonal, showing heavy underestimation of true severities. Predicted severities span a narrow range, while actual severities span 5+ orders of magnitude. Therefore, the model does not capture tail behavior: severity model is too conservative for large claims.

4.2.2 • TUNED MODEL

We use our above explained analysis of the model, to build a robust hyperparameter tuning in order to control frequency overestimation and improve tail modeling for severity. For frequency, we increased the minimum sample leafs, limited the maximum leaf nodes, and applied a stronger L2 regularization to enforce smoother,

more conservative rate estimates for rare events. As for severity, we progressively increased tree capacity and reduced the minimum samples leaf to better capture heavy-tailed losses, while keeping moderate regularization to avoid instability. We also lowered learning rates with kept sufficient boosting iterations to ensure stable convergence, and we tried multiple configurations to balance global calibration.

While we find "the best parameters", these parameters are wiggling the validation RMSE by noise-level amounts. The tuned configurations are essentially equivalent, all of them are clustered around 6777.2 - 6778.3, that's basically the same model family. Using the tuned model, we get a test $RMSE = 5600.20$, which is slightly (but not meaningfully) worse than our initial RMSE. This confirms our initial observation.

4.2.3 • MODIFIED MODEL

Since tuning our initial model was quite challenging, we try modifying it in hope to get a better RMSE, by using Tweedie loss for severity instead of Gamma. Here as well, we observe that the validation RMSE stays nearly constant ($RMSE = 6777.73$). We obtain a test $RMSE = 5600.58$. Using Tweedie loss for severity instead of Gamma, does not change the model's accuracy. A possible explanation would be that Gamma and Tweedie behave very similarly in the positive, right-skewed setting when trained with boosting and strong regularization, both losses converge to nearly the same fitted surface.

Observing that the model's performance remains almost constant when tuning the hyperparameters and changing the loss for severity, we hypothesize that this behaviour stems from the RMSE being largely driven by the big amount of near-zero charges combined with a few extreme claims that all tuned models handle in a very similar way. As a result, small changes in tree depth, regularization, or even the severity loss (Gamma vs Tweedie) do not materially alter the predicted charge distribution, so the metric barely moves.

4.2.4 • SYNTHESIS OF MODEL PERFORMANCE

The Histogram-based Gradient Boosting Regressor model delivers substantially better predictive structure than the GLM, effectively capturing non-linear relationships and ranking risks across deciles despite the extreme sparsity of the data. Frequency predictions show smooth and meaningful separation, and severity predictions better reflect the heavy-tailed distribution, though large losses remain underestimated. At the charge level, the model provides strong discrimination with moderate calibration gaps in the highest-risk decile. Combining accuracy, robustness to missing data, and fast training time, HGBR achieves the best overall performance and represents a practical and scalable solution for large insurance portfolios. Using this model, our team ranked 34th out of 340 participants, on the public leaderboard. Its computational efficiency, taking 2 minutes and 21 seconds for full training and prediction, makes it especially well suited to large-scale actuarial applications.

4.3 CATBOOST

CatBoost is a high-performance gradient boosting framework specifically optimized for datasets with a large number of categorical features. Unlike other boosting algorithms that require manual encoding (such as One-Hot or Ordinal encoding), CatBoost implements a proprietary Ordered Target Statistics algorithm to process categorical variables during training, significantly reducing overfitting and data leakage. Another structural distinction is the use of oblivious decision trees, which use the same splitting feature across an entire level of the tree. This symmetry leads to faster execution times during prediction and acts as a natural regularizer.

Our implementation follows the frequency-severity decomposition to maintain consistency with the actuarial framework:

- **Frequency Model:** Similar to the HGBR approach, claim counts are modeled using a Poisson loss function.
- **Severity Model:** For claim costs, we utilize the Tweedie loss function with a variance power of 1.5. This distribution is particularly effective as it serves as a compound model, bridging the gap between Poisson and Gamma distributions to better capture the heavy-tailed nature of agricultural fire claims. Training

is performed exclusively on the subset of contracts with strictly positive charges ($CHARGE > 0$), with costs clipped at a minimum of 1.0 to ensure numerical stability within the Tweedie framework.

Implementation and Performance: To ensure robustness, the models are regularized using a high `min_data_in_leaf` (150) and L2 regularization. While CatBoost provides superior accuracy on categorical data, its training phase is more computationally intensive than HGBR. On our dataset (383,610 rows), training an ensemble of 1,000 iterations requires more than 30 minutes, primarily due to the complex processing of categorical indices. To stabilize results, we utilize bagging across multiple random seeds. This approach yields precise results, successfully capturing the volatility of high-severity fire risks while maintaining a stable frequency calibration across all risk deciles.

4.3.1 • MODEL DIAGNOSTICS

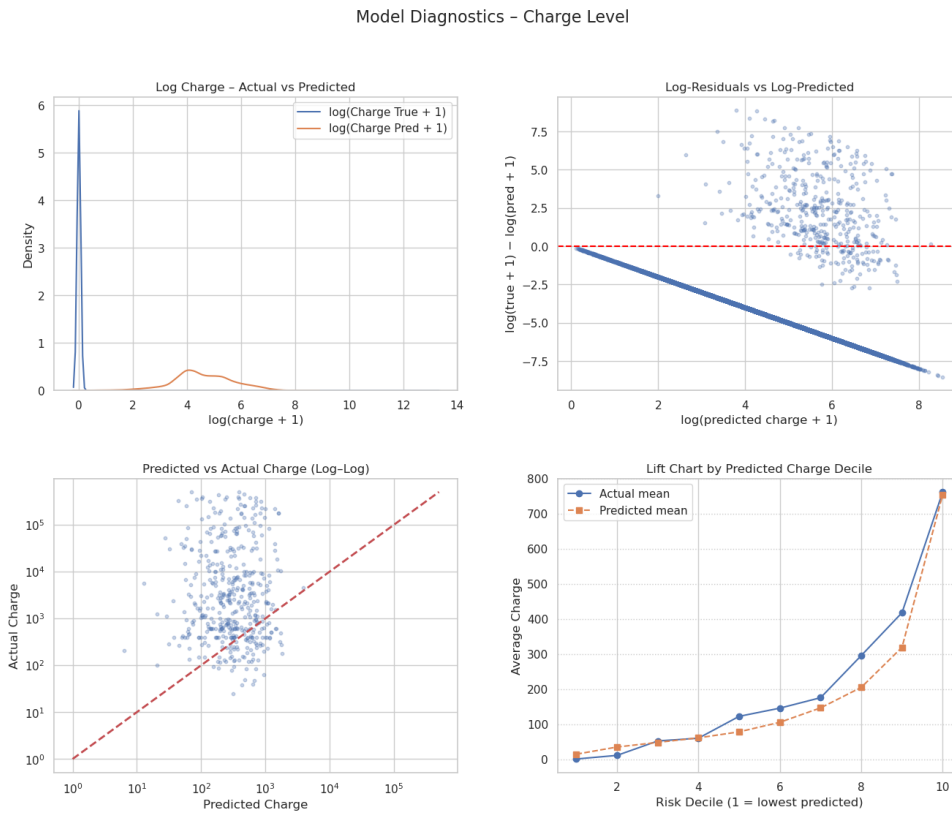


Figure 6: Model diagnostics at the Charge level using the CatBoost model

We analyse our model’s performance, based on the validation set predictions, at the charge level by plotting the following figures:

- **Log Charge – Actual vs Predicted:** This density plot compares the distribution of the logarithm of predicted versus actual charges. The model shows a smooth distribution of predictions centered around the mean, while the actual charges exhibit a significant mass at zero, highlighting the zero-inflation challenge of fire risks.
- **Log-Residuals vs Log-Predicted:** The residuals are centered around the zero-line for positive claims, indicating no major systematic bias in the logarithmic space. However, the diagonal boundary at the bottom reflects the lower limit of predictions for non-claiming contracts.

- **Predicted vs Actual Charge (Log-Log):** The scatter plot illustrates that while predictions are concentrated in a specific range, actual charges span a much wider interval. The alignment along the identity line for positive claims confirms that the model captures the order of magnitude of the risk, despite the inherent volatility.
- **Lift Chart by Predicted Charge Decile:** The model demonstrates strong discriminative power. The monotonic increase in both predicted and actual means across deciles shows that the model effectively identifies high-risk contracts, with the 10th decile accurately capturing the most expensive risks.

Model Diagnostics - Frequency & Severity

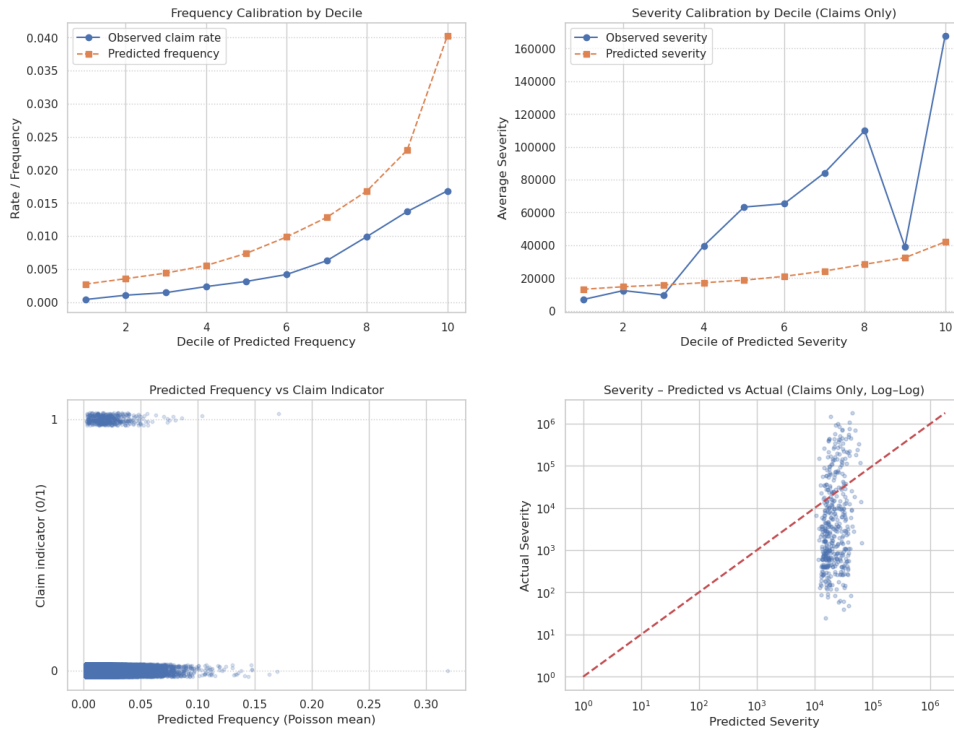


Figure 7: Model diagnostics at the Frequency and Severity level using the CatBoost model

We analyse our model’s performance, based on the validation set predictions, at the frequency and severity level by plotting the following figures:

- **Frequency Calibration by Decile:** The model captures the overall increasing trend in claim rates across deciles. Although there is a slight overestimation in the highest decile (D10), the monotonic upward pattern indicates a strong ability to rank contracts by their probability of claim occurrence.
- **Severity Calibration by Decile:** This plot reveals the complexity of severity modeling. While the predicted severity increases steadily across deciles, the observed severity exhibits substantial volatility, particularly in D7 and D9. The pronounced gap in D10 suggests that extreme, rare (“black swan”) fire events remain challenging to accurately quantify at an individual level.
- **Predicted Frequency vs Claim Indicator:** The scatter plot of the binary claim indicator against the predicted Poisson mean shows that higher predicted frequencies correspond to a greater concentration of observed claims, supporting the statistical relevance of the frequency model.

- **Severity – Predicted vs Actual (Claims Only, Log–Log):** On the subset of observed claims, predicted severities are broadly centered around the observed values. The large vertical dispersion reflects the heavy-tailed nature of fire losses, where similar risk profiles can lead to widely varying claim amounts.

4.3.2 • TUNED CATBOOST MODEL

Based on our diagnostic analysis, we performed a targeted hyperparameter tuning for the CatBoost framework to better control frequency calibration and capture the high-volatility tail of the severity distribution. For the frequency model, we applied a high L2 regularization (`l2_leaf_reg = 10`) to enforce more conservative rate estimates and mitigate the observed overestimation in high-risk deciles. For the severity model, we optimized the Tweedie variance power, identifying $p = 1.9$ as the optimal configuration. This higher power coefficient forces the model to place significantly more weight on extreme, heavy-tailed losses during the training phase.

Despite this granular optimization, the impact on global performance was marginal. The "best parameters" identified through GridSearch, specifically an `l2_leaf_reg` of 0.5 and a `variance_power` of 1.9 for severity, resulted in a test RMSE of 5599.58. This represents a negligible improvement over the non-tuned baseline RMSE of 5599.61. Such a minor variation (a difference of only 0.03) suggests that both the baseline and tuned configurations have converged to a nearly identical fitted surface.

4.3.3 • ANALYSIS OF TUNING STABILITY

The relative stability of the RMSE during tuning confirms our hypothesis that the metric is primarily driven by the extreme sparsity of the data, where the vast majority of charges are zero, and a few extreme claims that dominate the loss. In this context, small adjustments to regularization or the Tweedie power parameter do not materially alter the predicted charge distribution enough to shift the RMSE significantly.

While the improvement in accuracy was not statistically meaningful, the tuning process validated the robustness of our initial architecture. The CatBoost model proved particularly efficient at handling the high-dimensional categorical space of this challenge. However, this precision comes at a higher computational cost compared to HGBR; while the severity tuning on the claims-only subset was relatively fast (approximately 10 minutes), the frequency tuning on the full dataset required roughly 80 minutes to complete due to the complex Ordered Target Statistics processing. Ultimately, the CatBoost framework provides a stable actuarial model, though it remains constrained by the inherent volatility and "black swan" nature of agricultural fire risks.

4.3.4 • SYNTHESIS OF MODEL PERFORMANCE

In summary, the diagnostic analysis confirms that the CatBoost framework is well-suited for the complexities of agricultural fire risk. The model demonstrates high discriminative power, as evidenced by the sharp ascent of the Lift Chart, which successfully segregates high-risk profiles into the top deciles.

While the Severity Calibration reveals inherent volatility and a tendency to underestimate catastrophic "tail" events, this is partially compensated by a conservative Frequency estimation. At the aggregate level (Charge), the model remains well-calibrated, meaning that the total predicted premium for a large portfolio aligns closely with the observed losses. The residual analysis further confirms that our choice of the Tweedie loss function effectively handles the heteroscedasticity and zero-inflation present in the data. Overall, the model provides a robust and actuarially sound pricing structure that balances individual risk discrimination with aggregate financial stability.

5

MODEL SELECTION AND EVALUATION

Model comparison is based on two criteria: predictive accuracy, measured by test RMSE, and computational efficiency, measured as wall-clock time for training and prediction ³.

Model	GLM	HGBR	CatBoost
RMSE	5603.60	5598.75	5599.58
Time	1.6s	2min 21s	37min 30s

We observe that the baseline GLM runs significantly faster (1.6 seconds) than the tree-based models. Although the GLM achieves a seemingly competitive RMSE (5603.60), the diagnostic plots reveal that it effectively predicts near-constant values; this behavior goes largely unnoticed by RMSE because most observations have zero charge. In contrast, tree-based models capture far more structure in the data and therefore reach better accuracy, but at a higher computational cost. For that reason, while the GLM remains a meaningful benchmark and would be preferable if model simplicity were the main constraint, we ultimately favor tree-based methods.

Among these, both HGBR and CatBoost deliver similar performance (RMSE = 5598.75 and 5599.58, respectively). However, CatBoost is substantially more time-consuming (37 minutes 30 seconds versus 2 minutes 21 seconds for HGBR), which complicates both training and hyperparameter tuning. In practical insurance applications, where datasets are often much larger, computational efficiency becomes even more critical.

Considering the trade-off between predictive accuracy and computational cost, we select HGBR as our primary model: it achieves the best RMSE while remaining computationally manageable.

6

CONCLUSION

Pricing agricultural fire risk presents a challenging problem due to the extreme zero-inflation of claims and the heavy-tailed nature of catastrophic losses. In this study, we evaluated three modeling strategies: Generalized Linear Models (GLM), Histogram-based Gradient Boosting Regression (HGBR), and CatBoost, within a frequency–severity decomposition framework. Our findings underscore the importance of model flexibility, robustness to sparse and heterogeneous data, and computational efficiency.

The baseline GLM, while exceptionally fast to train, failed to capture meaningful variability in both frequency and severity components. As shown in the diagnostic plots, the model collapsed toward near-constant predictions, a limitation that RMSE alone does not reveal due to the dominance of zero charges. Tree-based methods provided a substantial improvement: both HGBR and CatBoost achieved far better discrimination across risk deciles and more realistic calibration of predicted premiums. CatBoost demonstrated strong performance on high-cardinality categorical features and produced the most stable risk ranking, but at a prohibitive computational cost, requiring more than 30 minutes per full run.

HGBR ultimately offered the best balance between accuracy and scalability. It achieved the lowest test RMSE (5598.75) while requiring only 2.3 minutes of runtime, making it substantially more efficient than CatBoost with no meaningful loss in predictive power. Given the size of the dataset, the presence of structured missingness, and the practical constraints of large-scale insurance modeling, HGBR stood out as the most operationally viable approach.

³All experiments were run on a Linux machine equipped with an Intel Xeon W-1290P CPU (10 physical cores, 20 threads, up to 5.3 GHz) and 62 GB of RAM, using Python 3.12.7. Runtime measurements correspond to wall-clock time and include both training and prediction phases.

Our 34th place ranking among 340 participants, on the public leaderboard, further validates the effectiveness of the frequency–severity decomposition, particularly when paired with gradient boosting techniques. Overall, we conclude that HGBR provides the optimal trade-off between predictive performance, interpretability, and computational feasibility for large-scale agricultural fire risk pricing.

REFERENCES

- [1] *Challenge Data*. Ecole Normale Supérieure. 2026. URL: <https://challengedata.ens.fr/>.
- [2] Dominik Chevalier and Marie-Pier Côté. “From point to probabilistic gradient boosting for claim frequency and severity prediction”. In: *arXiv preprint arXiv:2412.14916* (2025). URL: <https://arxiv.org/abs/2412.14916>.
- [3] Edward W. Frees. *Regression Modeling with Actuarial and Financial Applications*. Cambridge University Press, 2010.
- [4] Roel Henckaerts et al. “Boosting insights in insurance tariff plans with tree-based machine learning methods”. In: *arXiv preprint arXiv:1904.10890* (2019). URL: <https://arxiv.org/abs/1904.10890>.

APPENDIX

CODE

All code implemented for this project can be found here.